UVM Projects

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Project-1: CRC-8 Generator Verification using UVM

Project Description:

Developed a UVM-based verification environment for a CRC-8 generator designed using SystemVerilog. The design implements CRC calculation using polynomial 8'h07 ($x^8 + x^2 + x + 1$) with an initial CRC value of 8'h00. Developed a UVM testbench using object-oriented verification components, including Sequence Item, Sequence, Sequencer, Driver, Monitor, Agent, Scoreboard, Environment, and Test. Used transaction-based stimulus generation and scoreboard-based checking to verify the CRC output against expected results.

Key Features:

- CRC-8 verification using polynomial 8'h07.
- UVM-based testbench architecture.
- Sequence-based stimulus generation.
- Driver and Monitor communication through a virtual interface.
- Agent and Environment integration.
- Scoreboard-based functional checking.

Tools Used: SystemVerilog, UVM, EDA Playground.

Verification: UVM-based functional verification.

EDA Link: https://www.edaplayground.com/x/7V6v

Project-2: Traffic Light Controller Verification using UVM

Project Description:

Developed a UVM-based verification environment for a Traffic Light Controller designed using SystemVerilog and Finite State Machine (FSM) architecture. The controller consists of five states: RED, GREEN, YELLOW, PEDESTRIAN, and EMERGENCY, with configurable signal durations. Implemented pedestrian request handling with pending-request storage and emergency priority over normal traffic operations. Developed a UVM testbench using Transaction, Sequence, Sequencer, Driver, Monitor, Agent, Scoreboard, Environment, and Test components. Implemented an independent reference FSM in the scoreboard to compare expected and actual traffic signal outputs. Applied directed and constrained-random stimulus to verify pedestrian requests, emergency conditions, and traffic signal transitions.

Key Features:

- Five-state FSM-based traffic light controller.
- Configurable RED, GREEN, YELLOW, and pedestrian signal durations.
- Pedestrian request detection and pending-request handling.
- Emergency handling with priority over normal traffic operations.
- UVM transaction-based stimulus generation.
- Driver and Monitor communication through a virtual interface.
- Automated comparison of RED, GREEN, YELLOW, and WALK outputs.
- 20 transactions generated and monitored.

Tools Used: SystemVerilog, UVM, EDA Playground.

Verification: UVM-based functional verification using directed and constrained-random stimulus with scoreboard-based output checking.

EDA Link: https://edaplayground.com/x/BRUP

Project-3: 8b/10b Encoder and Decoder Verification using UVM

Project Description:

Developed an 8b/10b Encoder and Decoder using SystemVerilog and verified the design using UVM. The encoder converts 8-bit input data into a 10-bit encoded output, while the decoder converts the 10-bit data back into the original 8-bit data. The design supports K-characters and running disparity. Developed a UVM testbench to generate different input patterns and check the encoder and decoder outputs.

Key Features:

- Converts 8-bit data into 10-bit encoded data and decodes it back.
- Supports 5b/6b and 3b/4b encoding.
- Supports K-characters such as K28.5.
- Maintains running disparity during encoding and decoding.
- Detects invalid K-characters and invalid codewords.
- Uses directed and constrained-random test cases.
- Uses a monitor to collect and align encoder and decoder transactions.
- Uses a scoreboard to compare the expected and actual outputs.

Tools Used: SystemVerilog, UVM, EDA Playground.

Verification: Verified the encoder and decoder using directed and constrained-random inputs. Checked data recovery, K-character detection, valid signals, error signals, and running disparity using the UVM scoreboard.

EDA Link: https://www.edaplayground.com/x/pktZ

Project-4: Smart Vending Machine Controller Verification using UVM

Project Description:

Developed a Smart Vending Machine Controller using SystemVerilog and verified it using UVM. The controller uses a Finite State Machine (FSM) to accept coins, select products, check the available balance, and dispense the selected product. It also handles change calculation, cancellation, refunds, and invalid operations. Developed a UVM testbench to generate different transactions and verify the controller outputs.

Key Features:

- FSM-based vending machine controller.
- Accepts different coin values.
- Supports multiple product selections.
- Checks the balance before dispensing a product.
- Calculates and returns the remaining change.
- Supports transaction cancellation and refunds.
- Detects invalid operations.
- Uses a UVM scoreboard to compare expected and actual outputs.

Tools Used: SystemVerilog, UVM, EDA Playground.

Verification: Verified coin insertion, product selection, balance checking, product dispensing, change calculation, cancellation, and refund operations using UVM sequences and scoreboard-based checking.

EDA Link: https://www.edaplayground.com/x/qxbD